

AI BASED CONTEXTUAL ANSWER EVALUATION SYSTEM

PHASE 1 REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled " AI Based Contextual Answer Evaluation System " is the bonafide work of Roshini V (230701270), Sakshi P (230701279), Sandhiya P (230701282), Shradha S (230701311) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

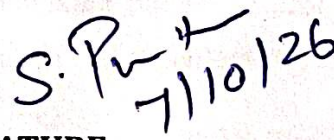
This project addresses the following Sustainable Development Goal:

Goal- No 4: Quality education



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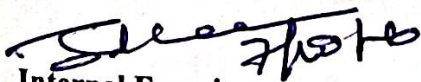
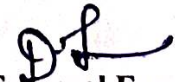
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ABSTRACT

The evaluation of descriptive answers is an important process in the educational system as it helps in assessing a student's understanding and knowledge of a subject. Traditional manual evaluation requires teachers to read and analyze each answer individually, making the process time-consuming and difficult when a large number of students are involved. Existing automated evaluation systems mainly depend on keyword matching and may not effectively understand answers expressed using different words or sentence structures.

The proposed **AI-Based Contextual Answer Evaluation System** is developed to assist teachers in evaluating digital descriptive answers using Artificial Intelligence, Natural Language Processing and Machine Learning techniques. The system analyses the contextual and semantic meaning of student answers by comparing them with reference answers using Sentence-BERT and cosine similarity. KeyBERT is used to identify important concepts, while rubric-based evaluation provides criterion-wise scoring. Machine Learning models such as Random Forest Regressor and XGBoost Regressor are used to support score prediction.

The system follows a teacher-in-the-loop approach where AI generates an initial evaluation and feedback, while the teacher reviews, modifies and approves the final result. This approach helps reduce evaluation time, improve consistency and provide meaningful feedback while maintaining human control over academic decisions. The proposed system aims to provide an efficient, transparent and reliable solution for contextual evaluation of descriptive answers.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The AI-Based Contextual Answer Evaluation System is developed to address the challenges associated with the manual evaluation of descriptive answers. Traditional answer evaluation requires significant time and effort from teachers and may result in variations in scoring and delayed feedback. The proposed system provides an AI-assisted approach that supports the evaluation process while maintaining the teacher as the final decision-maker.

The system integrates Natural Language Processing, Semantic Similarity Analysis, Concept Extraction, Rubric-Based Evaluation, and Machine Learning to analyze student answers. Sentence-BERT is used to understand the contextual similarity between the student answer and the reference answer, while KeyBERT is used to identify important concepts. The evaluation features are further processed using Random Forest Regressor and XGBoost Regressor for score prediction.

An important feature of the proposed system is the teacher-in-the-loop approach. AI-generated scores and feedback are reviewed by teachers before the final results are published. Teachers can modify scores, edit feedback, and approve or reject evaluations. This ensures that Artificial Intelligence assists the evaluation process without completely replacing human academic judgment.

The system also provides rubric-wise evaluation, personalized feedback, result management, analytics, and audit trail functionality. These features improve transparency and help students understand their performance and areas requiring improvement.

Thus, the proposed system provides an efficient and structured approach for evaluating digital descriptive answers. By combining contextual analysis, Machine Learning, and teacher supervision, the AI-Based Contextual Answer Evaluation System aims to reduce evaluation workload, improve consistency, and provide meaningful feedback while ensuring that final academic decisions remain under human control.